BILL OF MATERIALS

Robotics Platform - Hardware Components

Core Components

No.	Component	Quantity	Purpose
1	Genuine Arduino UNO R4 WiFi	1 pc	Main microcontroller and Wi-Fi enabled control board
2	L298N Motor Driver Module	1 pc	Dual H-bridge motor driver for DC motors
3	Breadboard Power Rail	1 pc	Power distribution
4	HC-SR04 Ultrasonic Sensor	1 pc	Distance and obstacle detection
5	IR Sensor	1 pc	Anti-falling / edge detection
6	MG90S Micro Servo	1 pc	Sensor positioning / steering mechanism
7	Ultrasonic Sensor Holder	1 pc	Mechanical mounting for HC-SR04
8	Rechargeable 18650 Batteries	2 pcs	Main battery supply
9	Double 18650 Battery Case	1 pc	Battery holder
10	25V Voltage Sensor Module	1 pc	Battery voltage monitoring
11	DC BO Motor - Single Shaft	4 pcs	4-wheel drivetrain
12	DC BO Motor Wheels	4 pcs	Wheels for BO motors
13	4WD Acrylic Chassis	1 set	Robot chassis and mounting platform
14	WS2812B RGB NeoPixel	1 pc	Programmable RGB status indicator
15	Normal / Hook Wire	3 m	Optional general-purpose wiring

Jumper Wires

Type	Quantity
Male-to-Male	20 pcs
Male-to-Female	20 pcs
Female-to-Female	20 pcs

Assembly Notes

The normal/hook wire is optional and may be used where additional cable length or permanent wiring is required. Battery wiring, polarity, motor-driver connections, and voltage limits should be checked carefully before powering the system.